

DIVYANSH NEGI

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EDUCATION

Birla Institute of Applied Sciences

B.Tech in Computer Science and Engineering

Bhimtal, Uttarakhand, India

August 2024 - Expected May 2028

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Software Engineering, Applied Artificial Intelligence

TECHNICAL SKILLS

- **Programming Languages:** Python, TypeScript, JavaScript, SQL, C++, Java, Bash
- **AI & Agent Engineering:** Google Agent Development Kit (ADK), FastMCP (Model Context Protocol), LangChain, Multi-Turn ReAct Loops, Tool / Function Calling, Self-Healing Reflection Loops, RAG, Vector Search, LLM Evaluation Harnesses (Faithfulness, MRR)
- **Backend & Cloud Platforms:** FastAPI, Node.js, Express, Docker, CI/CD (GitHub Actions), REST & GraphQL APIs, Server-Sent Events (SSE), GCP, OCI, Linux/Unix
- **Databases & Vector Stores:** ClickHouse Cloud, PostgreSQL, DuckDB, ChromaDB, Pinecone, MongoDB, Redis

PROJECTS

OmniQuery AI — Autonomous Film Analytics & Agentic Platform

[GitHub](#) | [Live Demo](#)

Google ADK, Gemini 3.6 Flash, ClickHouse Cloud (GCP), FastMCP, DuckDB, FastAPI, React 19, Recharts

- **Architected** an autonomous conversational analytics agent using Google Agent Development Kit (ADK) and Gemini 3.6 Flash, executing multi-turn schema introspection, dynamic query planning, and sub-200ms OLAP queries across 50,000+ records.
- **Standardized** agent tool-use via Model Context Protocol (mcp-clickhouse / FastMCP), orchestrating a self-healing SQL reflection loop that catches dialect syntax exceptions to regenerate valid queries with a 99%+ execution success rate.
- **Implemented** 16-dimensional cosine similarity vector search over 5,000+ audience reviews alongside an automated dual-engine fallback from ClickHouse Cloud on GCP to in-memory DuckDB, preventing tool execution outages.
- **Engineered** real-time Server-Sent Events (SSE) streaming of agent reasoning traces, intermediate tool logs, and KPI visual payloads to a responsive React 19 and Recharts dashboard.

Full-Stack AI Document Intelligence & RAG Platform

[GitHub](#)

FastAPI, Express.js, React, ChromaDB, Google Gemini, Groq (LLaMA 3.3), PyMuPDF, EasyOCR

- **Designed** a distributed microservices agent pipeline with an Express.js gateway and FastAPI worker, implementing an LLM query classifier (Gemini Flash) that dynamically routes between single-hop extraction and multi-hop synthesis to cut token costs by 35%.
- **Constructed** a resilient multi-stage document ingestion harness combining PyMuPDF digital extraction with EasyOCR image fallback, chunking and indexing text into ChromaDB for high-precision semantic search.
- **Benchmarked** retrieval faithfulness and latency across LLM providers, implementing real-time streaming failover (Gemini 1.5 Flash to Groq LLaMA 3.3 70B) to guarantee zero-downtime conversational query delivery.

AI-Based NLP Legal Summarization & Evaluation Engine

[GitHub](#)

Python, LangChain, HuggingFace Transformers, ChromaDB, RAG, Multi-Agent Evaluation

- **Engineered** an automated NLP contract distillation pipeline with LangChain and HuggingFace Transformers, condensing dense 50+ page legal agreements into structured 1-page executive briefs.
- **Optimized** semantic chunking and dense vector indexing in ChromaDB, increasing relevant clause retrieval accuracy by 40% while reducing search latency across document corpuses.
- **Orchestrated** an autonomous multi-agent evaluation harness to systematically score output coherence, summary completeness, and factual faithfulness against human benchmark contracts.

EXPERIENCE & TRAINING

Software Engineering Trainee / Learner | Full Stack Open (University of Helsinki)

Online

January 2026 - September 2026

- **Architected** scalable REST and GraphQL APIs with Node.js, Express, and Apollo Server, designing Mongoose schemas to resolve data over-fetching and decrease client payload sizes by 30%.
- **Automated** comprehensive integration test suites using Jest and Supertest in a GitHub Actions CI/CD pipeline, achieving 90%+ code coverage and preventing deployment regressions.
- **Constructed** type-safe React and TypeScript interface modules with Tailwind CSS, achieving WCAG 2.1 AA accessibility compliance and sub-second client render times.

CERTIFICATIONS & AWARDS

• **Full Stack Open: Containers (Docker)**, University of Helsinki

[Verify](#) | Sep 2026

• **Full Stack Open: TypeScript**, University of Helsinki

[Verify](#) | Aug 2026

• **LLM Zoomcamp 2026**, DataTalks.Club (Credential ID: 6F7D81)

[Verify](#) | Aug 2026

• **Full Stack Open: GraphQL**, University of Helsinki

[Verify](#) | Jul 2026

• **Full Stack Open: Core (React, Node, CI/CD)**, University of Helsinki

[Verify](#) | Jun 2026

• **Applied Data Science Lab**, WorldQuant University

[Verify](#) | Apr 2026

• **Oracle Cloud Infrastructure 2025 AI Foundations Associate**, Oracle (ID: 323406956OCI25AICFA)

[Verify](#) | Oct 2025